Gaston Elian Bujia

Data Scientist | Machine Learning | Predictive Modeling | Python | Computational Statistics

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Industry Profile

Data Scientist with a strong background in mathematics, machine learning, and applied research. I combine hands-on experience building and shipping predictive models, including a production-deployed clinical scoring system, with doctoral training in computational modeling, Bayesian methods, and deep learning. I am especially interested in roles where rigorous quantitative thinking, Python-based analysis, and clear communication can turn complex data into actionable decisions.

Core Skills

- **Python and ML stack:** Python, NumPy, Pandas, scikit-learn, XGBoost, PyTorch, HuggingFace Transformers, Weights & Biases, SciPy, Jupyter Notebooks.
- **Predictive modeling:** Deep learning, NLP/audio models, model evaluation, clinical scoring, disease progression monitoring, and low-data prediction problems.
- **Statistics:** Computational statistics, Bayesian modeling, linear mixed-effects models, survival analysis with lifelines and statsmodels, Kaplan-Meier analysis, Cox proportional hazards models.
- **Visualization and communication:** Matplotlib, Seaborn, Plotly, technical writing, teaching, and communication of quantitative results to technical audiences.
- **Tools:** Git/GitHub, Docker containerization basics, LaTeX, open research software, and collaborative technical workflows.
- **Languages:** Spanish native; English professional working proficiency.

Relevant Experience

Research Data Scientist Everything ALS | *March 2024 - Present*

- Developed a dysarthria severity scoring model using Whisper (HuggingFace Transformers), PyTorch, and scikit-learn; the model is deployed in production and integrated into clinical trial workflows.
- Built CNN and transformer-based predictive models on accelerometer and voice signals for disease progression monitoring in ALS patients.
- Applied survival analysis with Kaplan-Meier curves and Cox proportional hazards models using lifelines and statsmodels to study medication impact and time-to-event outcomes.
- Collaborated directly with clinical scientists to translate research requirements into ML solutions for complex clinical datasets.

Graduate Researcher Applied Artificial Intelligence Lab (UBA-CONICET) | *2018 - Present*

- Designed and evaluated end-to-end ML pipelines for predicting sequential human behavior from real experimental datasets, combining deep learning, Bayesian modeling, CNN-based saliency models, and reproducible analysis workflows.
- Contributed to **ViSioNS** (NeurIPS 2022), an open-source benchmark for evaluating computational models, analogous to model evaluation frameworks used in production ML.

- Developed **sIBS**, a probabilistic visual-search model combining Bayesian inference and saliency maps, published in *Frontiers in Systems Neuroscience*.

Technical Teaching and Mentoring

Senior Teaching Assistant / Teaching Assistant **Department of Computer Science, School of Exact and Natural Sciences, University of Buenos Aires** | *2021 - Present*

- University teaching in **Machine Learning**, **Computational Statistics**, and computer systems for undergraduate and graduate audiences.
- Experience translating quantitative and computational concepts into clear, structured technical material.

Thesis Co-supervisor and Undergraduate Research Mentor **University of Buenos Aires** | *2021 - Present*

- Mentored undergraduate research and thesis projects involving computational modeling, optimization, and behavioral data.
- Guided junior collaborators through problem formulation, implementation choices, and technical communication.

Key Projects

- **Dysarthria scoring model (Everything ALS):** Production-deployed clinical NLP/audio model built with Whisper, PyTorch, and scikit-learn.
- **ViSioNS (NeurIPS 2022):** Open-source benchmark for computational models of human visual search, adopted by the research community.
- **Recommendation systems (B.Sc. thesis):** Implemented Factorization Machines to address the cold-start problem in low-data recommendation scenarios.

Education

- **Ph.D. in Computer Science** | UBA (2018 - 2026)
  - Doctoral thesis titled *Computational aspects of human vision: prediction of eye movements during visual search*. Expected defense between July and August 2026. Advisor: Dr. Juan Kamienkowski. Co-advisor: Dr. Guillermo Solovey.
- **B.Sc. in Mathematics** | UBA (2018)
  - Applied mathematics focus. Thesis: *Recommendation systems: Factorization Machines and the cold-start problem.* Advisor: Dr. Leandro Lombardi.
  - Relevant to recommender systems, embeddings, and ranking problems.

Selected Research Output

1. Ruarte, G., **Bujia, G.**, Care, D., Ison, M. J., & Kamienkowski, J. E. (2025). *Integrating Bayesian and neural networks models for eye movement prediction in hybrid search.* In **Scientific Reports**, 15(1), 16482.
2. Dziemian, S., **Bujia, G.**, Prasse, P., Barańczuk-Turska, Z., Jäger, L. A., Kamienkowski, J. E., & Langer, N. (2025). *Saliency models reveal reduced top-down attention in Attention-Deficit/Hyperactivity Disorder: A naturalistic eye-tracking study.* In **JAACAP Open**, 3(2), 192–204.
3. **Bujia, G.**, Ruarte, G., Sclar, M., Solovey, G., & Kamienkowski, J. E. (2025). *Uncertainty during visual search: Insights from a computational model and behavioral experiment in natural stimuli.* In **bioRxiv**.